

# — CEVAP ANAHTARI —

|                                                                                                                                                                                                                |                                                                                            |  |                   |      |               |            |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--|-------------------|------|---------------|------------|------------------|
|                                                                                                                                                                                                                | <b>YTU – Faculty of Naval Architecture and Maritime, Final Exam, Questions and Answers</b> |  | <b>NOTE CHART</b> |      |               |            |                  |
|                                                                                                                                                                                                                |                                                                                            |  | 1. q              | 2. q | 3. q          | 4. q       | Total            |
| Student Name and Surname                                                                                                                                                                                       |                                                                                            |  |                   |      |               |            |                  |
| Students Number                                                                                                                                                                                                |                                                                                            |  |                   |      |               |            |                  |
| Section                                                                                                                                                                                                        | Mathematical Engineering                                                                   |  |                   |      | Exam Date     | 09/11/2016 |                  |
| Course Name                                                                                                                                                                                                    | Linear Algebra                                                                             |  | Group Number      |      | Exam Duration | 80         | Examination Room |
| Course Instructor Name and Surname                                                                                                                                                                             | FATMA AYDIN AKGUN                                                                          |  |                   |      | Signature     |            |                  |
| Student Disciplinary Regulations "and to make or attempt to make copies of exams to" the actual perpetrators are suspended from one or two semesters. (YÖK; 2547 Student Disciplinary Regulations, 9. Article) |                                                                                            |  |                   |      |               |            |                  |

1) In  $\mathbb{R}^3$ , a) Show that the vectors  $u_1 = (1, -1, 0)$ ,  $u_2 = (1, 0, 2)$ ,  $u_3 = (1, 2, 1)$  are linearly independent.

b) Rewrite the vector  $u = (3, 5, 4)$  as linear combination of the vectors  $u_1, u_2, u_3$ . (20p)

2) If  $A^{-1} = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & 3 \\ -1 & -1 & -1 \end{bmatrix}$  and  $A^{-1}B = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}$  are given.

Write the linear equation system  $AX = B$ . (30p)

3) 
$$\begin{cases} x + 2y - 3z = 4 \\ 3x - y + 5z = 2 \\ 4x + y + (a^2 - 14)z = a + 2 \end{cases}$$
 Determine the value(s) of  $a$ , such that the equation system has

(25p) a) unique solution  
b) infinite solution  
c) no solution (inconsistent)

4) In  $V = P_2$  Euclid Space  $p(t) = t + 2$ ,  $q(t) = 2t - 3$  are given. If the inner product in  $V$  is defined as

$$\langle p(t), q(t) \rangle = \int_0^1 p(t) \cdot q(t) dt,$$

obtain the angle between  $p(t)$  and  $q(t)$ . (25p)

20 puan.

(1) For linear independency  
 $a_1 u_1 + a_2 u_2 + a_3 u_3 = 0 \Leftrightarrow a_1 = a_2 = a_3 = 0$  (Two ways)

a) 
$$\begin{cases} a_1 + a_2 + a_3 = 0 \\ -a_1 + 2a_3 = 0 \\ 2a_2 + a_3 = 0 \end{cases}$$

(2) 
$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix} \xrightarrow{R_2 + R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 1 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$$

(3) 
$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix} \xrightarrow{R_1 - R_2} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix} \xrightarrow{R_1 \cdot (-1)} \begin{bmatrix} -1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix} \xrightarrow{R_1 \cdot (-1)} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$$

Homogeneous system  $|A| \neq 0$  trivial solution  $(0, 0, 0)$

$a_1 = a_2 = a_3 = 0$  (3) linearly independent

Good Luck..

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix} \xrightarrow{r_2(1)} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 1 \end{bmatrix} \xrightarrow{r_3(1-2)} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix} \xrightarrow{r_1(-1)} \begin{bmatrix} 0 & 0 & -3 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix} \xrightarrow{r_1(3/5)} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix} \xrightarrow{r_2(-3)} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -5 \end{bmatrix} \xrightarrow{r_3(1/5)} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \xrightarrow{r_3(-1)} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{r_3(-1)} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$r_2(1)$   $r_3(1-2)$   $r_3(1/5)$   $r_1(-1)$   $r_1(3/5)$   $r_2(-3)$   $r_3(1/5)$   $r_3(-1)$   $r_3(-1)$

linearly independent

$$b) \quad a_1 v_1 + a_2 v_2 + a_3 v_3 = (3, 5, 4)$$

$$a_1 + a_2 + a_3 = 3$$

$$-a_1 + 2a_3 = 5$$

$$2a_2 + a_3 = 4$$

$$\begin{bmatrix} 1 & 1 & 1 & | & 3 \\ -1 & 0 & 2 & | & 5 \\ 0 & 2 & 1 & | & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & | & 3 \\ 0 & 1 & 3 & | & 8 \\ 0 & 2 & 1 & | & 4 \end{bmatrix}$$

$r_{21}(-1)$   $r_{12}(-1)$   
 $r_{32}(-2)$

$$\sim \begin{bmatrix} 1 & 0 & -2 & | & -5 \\ 0 & 1 & 3 & | & 8 \\ 0 & 0 & -5 & | & -12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & | & -5 \\ 0 & 1 & 3 & | & 8 \\ 0 & 0 & 1 & | & 12/5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & | & -11/5 \\ 0 & 1 & 0 & | & 4/5 \\ 0 & 0 & 1 & | & 12/5 \end{bmatrix}$$

$r_{31}(-1/5)$   $r_{23}(-3)$   
 $r_{13}(2)$

$$-v_1 + 4v_2 + 2v_3 = -11/5$$

$$a_1 = -11/5, a_2 = 4/5, a_3 = 12/5$$

3. par.

$$(2) \quad |A^{-1}| = \begin{vmatrix} 1 & 2 & 2 \\ 1 & 2 & 3 \\ -1 & -1 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 2 \\ 1 & 0 & 3 \\ -1 & 1 & -1 \end{vmatrix} = -1 \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = -1 \neq 0.$$

$$(2) \quad A = (A^{-1})^{-1} \Rightarrow (A^{-1})^t = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 3 & -1 \end{bmatrix} \quad (A^{-1})^* = \begin{bmatrix} 1 & 0 & 2 \\ -2 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$A = (A^{-1})^{-1} = \frac{A^*}{|A|} = \begin{bmatrix} -1 & 0 & -2 \\ 2 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$A^{-1}B = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \quad A \cdot A^{-1}B = B = \begin{bmatrix} -1 & 0 & -2 \\ 2 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix}$$

$$AX = B \quad \begin{bmatrix} -1 & 0 & -2 \\ 2 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix} //$$

3) 
$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 3 & -1 & 5 & 2 \\ 4 & 1 & (a^2-14) & a+2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & -7 & a^2-2 & a-14 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & 0 & a^2-16 & a-4 \end{bmatrix}$$

$r_2 \cdot (-3)$        $r_{32} \cdot (-1)$        $\textcircled{10}$

$r_3 \cdot (-4)$

$$a^2 - 16 = 0 \quad a = \pm 4 \quad a - 4 = 0 \quad a = 4$$

1)  $a = 4$      $r_A = r_B = 2 < n = 3$  infinite solution  $\textcircled{5}$

2)  $a = -4$      $r_A = 2$      $r_B = 3$      $r_A \neq r_B$  no solution  $\textcircled{5}$

3)  $a \neq \pm 4$      $r_A = r_B = n = 3$  unique solution.  $\textcircled{5}$

25 pun.

4)  $\cos \Theta = \frac{\langle p(t), q(t) \rangle}{\|p(t)\| \|q(t)\|}$

$$\begin{aligned} \langle p(t), q(t) \rangle &= \int_0^1 (t+2)(2t-3) dt = \int_0^1 (2t^2 + t - 6) dt = \left[ \frac{2t^3}{3} + \frac{t^2}{2} - 6t \right]_0^1 \\ &= \frac{2}{3} + \frac{1}{2} - 6 \\ &= \frac{7}{6} - 6 = \frac{-29}{6} \end{aligned} \quad \textcircled{7}$$

$$\|p(t)\| = \sqrt{\langle p(t), p(t) \rangle}$$

$$\begin{aligned} &= \int_0^1 (t+2)^2 dt = \int_0^1 (t^2 + 4t + 4) dt = \left[ \frac{t^3}{3} + 2t^2 + 4t \right]_0^1 \\ &= \frac{1}{3} + 2 + 4 = \frac{19}{3} \end{aligned} \quad \textcircled{7}$$

$$\begin{aligned} \|q(t)\| &= \sqrt{\langle q(t), q(t) \rangle} = \sqrt{\int_0^1 (2t-3)^2 dt} = \sqrt{\int_0^1 (4t^2 - 12t + 9) dt} \\ &= \sqrt{\left[ \frac{4t^3}{3} - 6t^2 + 9t \right]_0^1} = \sqrt{\frac{4}{3} - 6 + 9} = \sqrt{\frac{13}{3}} \end{aligned} \quad \textcircled{7}$$

4)  $\cos \Theta = \frac{-29}{6} \cdot \frac{1}{\sqrt{19 \cdot 13}} = \frac{-29}{2\sqrt{19 \cdot 13}}$        $\Theta = \arccos\left(\frac{-29}{2\sqrt{19 \cdot 13}}\right) //$